

WEBSITE TRAFFIC ANALYSIS REPORT

Alfido Tech Internship

Submitted By

Name: Sai Nikhila Bharatula

College: Chaitanya Bharathi Institute of Technology

Department: Artificial Intelligence and Data Science

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1. Objective

The main objective of this project was to analyze website traffic data to understand user activity and identify meaningful traffic patterns. By cleaning the dataset, exploring different event types, and creating visualizations, I aimed to study how website traffic varied across different countries, cities, artists, albums, and links. The insights obtained from this analysis can help improve website performance and support better business decisions.

2. Dataset Description

For this project, I used the **Website Traffic Analysis Dataset** available on Kaggle. The dataset contains over **226,000 website traffic events** collected from user interactions with music-related links.

The dataset includes the following important attributes:

- Event
- Date
- Country
- City
- Artist
- Album
- Track
- ISRC
- Link ID

The dataset provides useful information about website traffic events, geographical distribution, and content interactions. Although it does not contain user sessions or session duration, it offers valuable insights into website activity and visitor behaviour.

3. Data Cleaning and Preprocessing

Before starting the analysis, I carefully examined the dataset to understand its structure and quality.

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked the number of rows and columns.
- Verified column names.
- Examined data types.
- Checked for missing values.
- Checked for duplicate records.
- Converted the Date column into datetime format.
- Prepared the dataset for further analysis.

The dataset required only minor preprocessing before analysis. Most of the work involved checking data quality, handling the date column correctly, and preparing the dataset for visualization.

4. Feature Engineering

To improve the analysis, additional features were created from the existing Date column.

The **Year**, **Month**, and **Day** columns were extracted to make it easier to analyze website activity over time.

These additional columns simplified the analysis and helped create time-based visualizations.

5. Exploratory Data Analysis

After cleaning the dataset, I explored it using summary statistics and different visualizations.

Since most of the columns contained categorical values, the analysis mainly focused on identifying the most frequently occurring values and understanding traffic distribution.

The exploratory analysis mainly focused on:

- Website events
- Country-wise traffic
- City-wise traffic
- Daily website activity
- Artist popularity
- Album popularity
- Link activity

Exploring the dataset before creating visualizations helped me better understand the traffic patterns and identify meaningful trends.

6. Key Performance Indicators (KPIs)

The following KPIs were calculated using the available dataset:

- Total Website Events
- Total Unique Links
- Total Countries
- Total Cities

These KPIs provide a simple overview of website activity and geographical coverage.

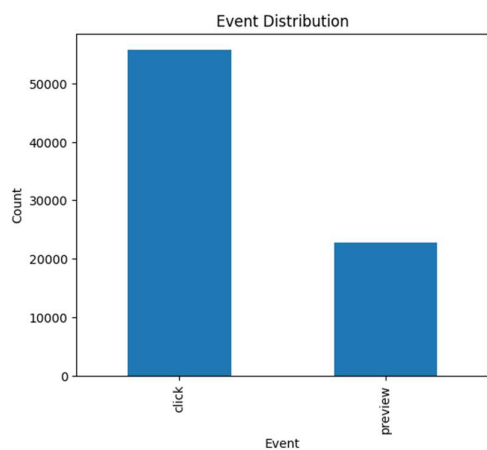
Note: Session IDs, User IDs, Bounce Rate, and Average Session Duration could not be calculated because these fields are not available in the provided dataset.

7. Data Visualization

Several visualizations were created to better understand website traffic patterns.

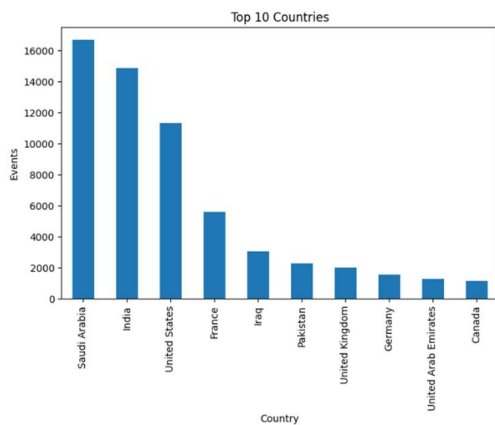
Event Distribution

A bar chart was created to compare different types of website events. This helped identify which events occurred most frequently.



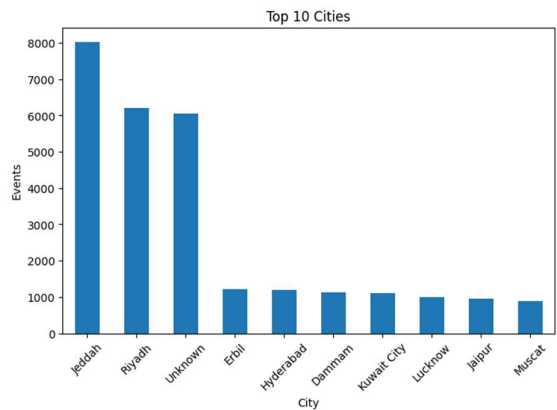
Top 10 Countries

This visualization shows the countries generating the highest number of website events and highlights geographical traffic distribution.



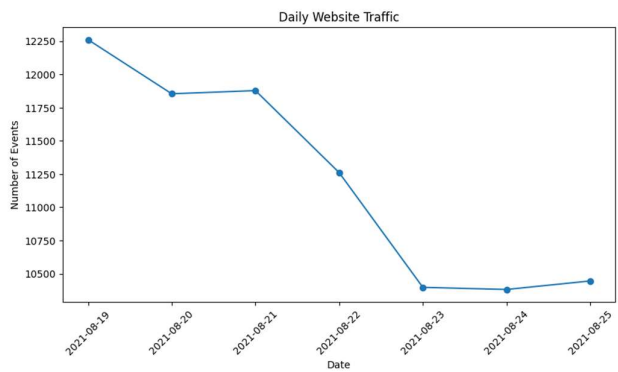
Top 10 Cities

A bar chart was created to compare website traffic across different cities. It provides a better understanding of locations with higher user activity.



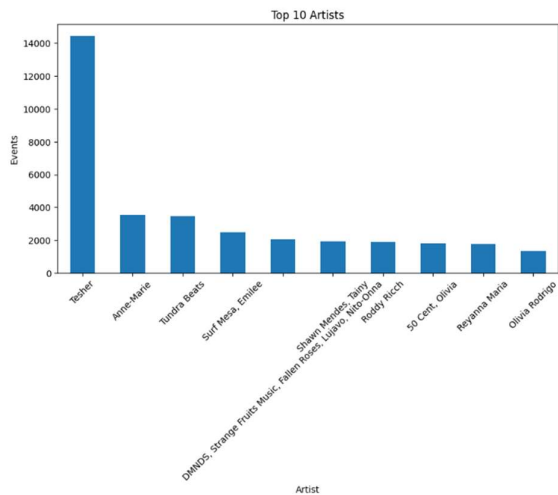
Daily Website Traffic

A line chart was created to observe how website activity changed over different dates. This visualization helps identify variations in daily traffic.



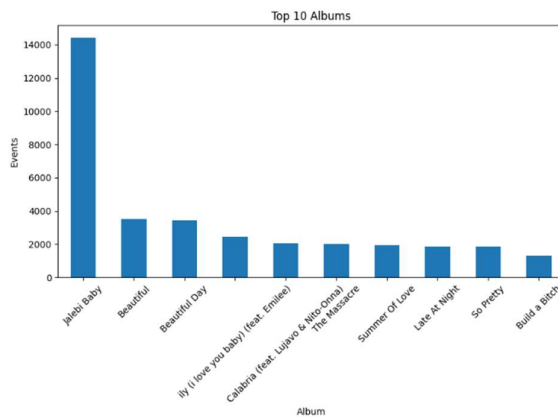
Top 10 Artists

This visualization compares the artists receiving the highest number of website events, helping identify the most popular content.



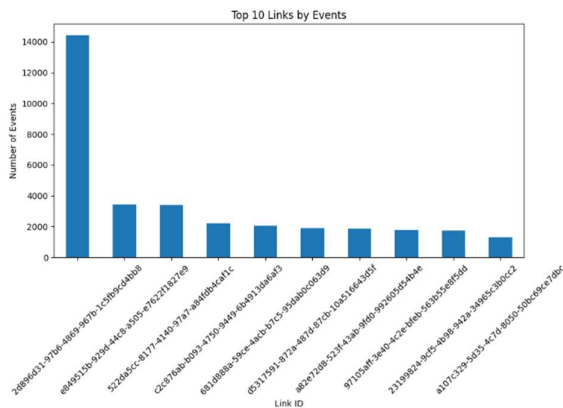
Top 10 Albums

A bar chart was created to compare albums based on the number of website events. It highlights the albums that attracted greater user interest.



Top 10 Links by Events

This chart shows the links with the highest number of recorded website events. It provides useful insights into the most frequently accessed content.



8. Website Traffic Analysis

After analysing the dataset, I observed that website activity was not evenly distributed across all countries, cities, artists, and links. Some locations generated significantly more traffic than others, while a few artists, albums, and links attracted a much higher number of events.

The KPI analysis also provided a useful summary of overall website activity by showing the total number of events, unique links, countries, and cities represented in the dataset. Together with the visualizations, these metrics helped provide a better understanding of website traffic behaviour.

9. Key Findings

After analysing the dataset and the visualizations, I observed several important patterns.

- The dataset contains more than 226,000 website traffic events.
- Website traffic was distributed across multiple countries and cities.
- Some artists and albums attracted considerably higher user activity.

- A small number of links generated a large share of total events.
 - The KPI summary provided a clear overview of website activity and geographical coverage.
 - Daily traffic showed variations over different dates.
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10. Recommendations

Based on the analysis, I believe the following improvements could help increase website engagement and improve user experience.

- Promote the most popular links to increase user engagement.
 - Improve content recommendations based on popular artists and albums.
 - Focus marketing efforts on countries and cities generating higher traffic.
 - Regularly monitor website traffic trends to identify changes in user behaviour.
 - Analyze high-performing content to improve future promotional strategies.
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11. Conclusion

Working on this project gave me practical experience in analysing website traffic data using Python.

During this project, I learned how to clean event-based datasets, perform exploratory data analysis, create meaningful visualizations, and interpret website traffic patterns. The project also helped me understand how traffic data can be used to support better business decisions.

Overall, this project improved my analytical skills and gave me more confidence in working with real-world website traffic datasets.